constrained by weak radio signals and small fuel capacities, restricting operations to short, line-of-sight distances.

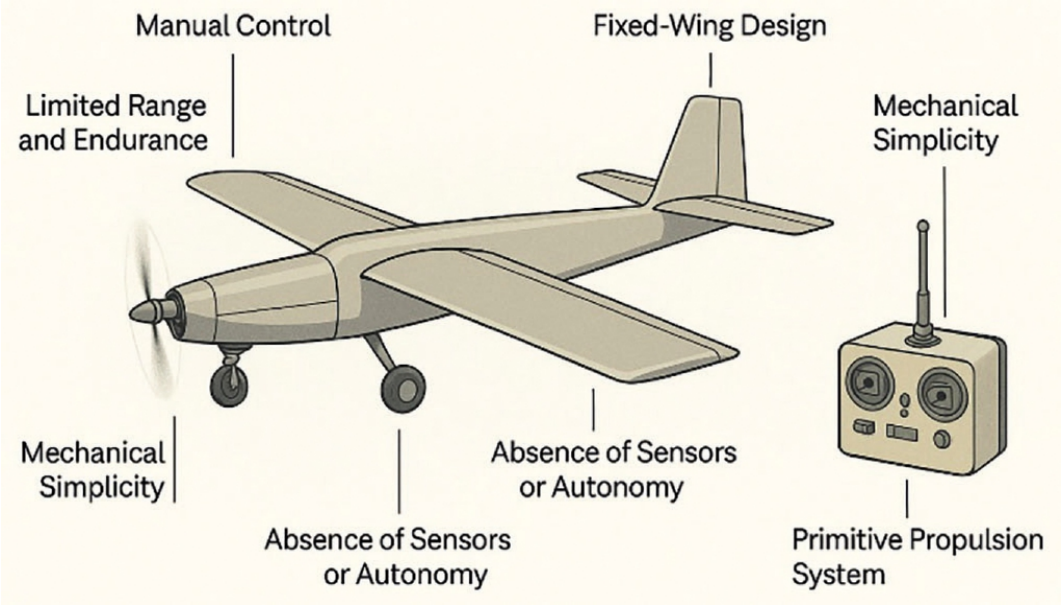


Figure 2.1 Basic characteristics of first-generation drones.⏎

The illustration also emphasizes the mechanical simplicity of first-generation drones – lacking sensors, automation, or feedback systems. Control surfaces such as rudders and elevators were managed via direct servo linkages, while propulsion depended on primitive piston or combustion engines. The absence of sensors or autonomous systems made flight stability and navigation fully dependent on the operator's visual tracking and skill.

2.2 Early Concepts of Unmanned Aerial Vehicles (UAVs)

The early concepts of unmanned aerial vehicles (UAVs) emerged from humanity's enduring desire to achieve controlled flight without risking human lives. Long before drones became technologically feasible, the idea of remotely guided or pilotless aircraft had already taken shape through both military vision and scientific experimentation. These early concepts